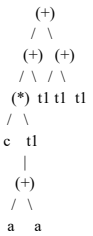
Explain *code optimization techniques in detail*.
Code optimization is the process of improving the intermediate code or target code to make it **more efficient** without changing its output. It improves: >Execution speed //>Memory usage >Power efficiency
Objectives of Code Optimization: 1. Reduce execution time //2. Minimize memory usage // 3.Eliminate redundant computations 4.Improve overall program performance
Types of Code Optimization
1.Machine Independent // 2.Machine Dependent Optimization
1. Machine Independent Optimization
These techniques do not depend on machine architecture.
Important Techniques: 1. Constant Folding: Compile-time evaluation of constant expressions. // Example: // a = 5 + 3 → a = 8
2. Constant Propagation: Replace variables having constant values. // Example: // a = 10 // b = a + 5 → b = 10 + 5
3. Common Subexpression Elimination (CSE): Avoid recomputing same expression. // Example:
x = a + b // y = a + b // → y = x
4. Dead Code Elimination: Removes unused or unreachable code. // Example: // x = 10 // x = 20 (first statement is useless)
5. Loop Optimization: Improves performance of loops.
(a) Loop Invariant Code Motion: Moves constant expressions outside loop. // Example: // for(i=0;i<n;i++){// x = a + b // }
→ x = a + b // for(i=0;i<n;i++){}
(b) Strength Reduction: Replace costly operations with cheaper ones. // Example: // x = y * 2 → x = y + y

Explain *runtime environment of compiler*.
>> The **runtime environment** of a compiler refers to the **environment in which a compiled program executes**.
It includes the **organization of memory, storage allocation, and management of variables and functions during program execution**.
It is responsible for **efficient execution of the target program**.
Functions of Runtime Environment:1. Allocation of memory 2.Management of function calls 3.Handling of variables (local & global) 4.Parameter passing 5.Supporting recursion 6.Maintaining execution order
Components of Runtime Environment
1. Storage Allocation
It deals with how memory is assigned to variables and functions.
Types of Storage Allocation
(a) Static Allocation: Memory allocated at **compile time**
> Fixed size // Example: Global variables// Fast access
No dynamic behavior
(b) Stack Allocation: Memory allocated at **runtime using stack**
Used for function calls
Supports recursion: Efficient // ❌ Limited size
(c) Heap Allocation: Memory allocated **dynamically at runtime**
>> Used for dynamic structures
Example: malloc(), new // ✓ Flexible // ❌ Slower and complex
2. Activation Record (Stack Frame)
An **activation record** is a block of memory used to store information about a function call.

Explain *lexical analysis in detail* with token recognition.
Lexical Analysis is the **first phase of a compiler**.
It reads the source program as a sequence of characters and converts it into a sequence of **tokens**. // It is also called a **scanner**.
Functions of Lexical Analyzer:1. Converts input program into **tokens** 2.Removes **whitespace and comments** 3. Identifies **lexical errors** 4. Creates entries in **symbol table** 5. Passes tokens to the parser
Basic Terminology// Token: A token is a **valid unit** of a program.
Example: // int a = 10; // → Tokens: int, a, =, 10
Lexeme: Actual sequence of characters forming a token.
Example: /// Token: identifier // Lexeme: a
Pattern: Rule used to define a token.
Example:// Identifier → [a-zA-Z][a-zA-Z0-9]*
Process of Lexical Analysis
Source Code → Lexical Analyzer → Tokens → Syntax Analyzer
Token Recognition: Token recognition is the process of identifying tokens from input using **patterns and finite automata**.
Techniques Used: 1. Regular Expressions: Used to define patterns for tokens //Examples: // Identifier → letter (letter | digit)*
Number → digit+
2. Finite Automata (DFA): Used to recognize tokens efficiently // Converts regular expressions into **DFA**
Example for identifier: // Start → letter → (letter/digit loop)
Example of Token Recognition
Input: // a = b + 10

Consider the expression:
(c(a + a) + (a + a)) + ((a + a) + (a + a))
Construct the Directed Acyclic Graph (DAG) for it.
Step 1: Identify Common Subexpressions
The expression **(a + a)** appears multiple times
So we create **only ONE node** for: // (a + a)
Let: // t1 = a + a
Step 2: Rewrite Expression: // (c * t1 + t1) + (t1 + t1)
Step 3: Create DAG Nodes: Leaf Nodes: // a // c
Internal Nodes: 1. t1 = a + a // 2. **t2 = c * t1** // 3. **t3 = t2 + t1** // 4. **t4 = t1 + t1** // 5.**Final = t3 + t4**
DAG Representation



Construct *SLR parsing table* for given grammar.
SLR (Simple LR) parser is a type of **bottom-up parser** that uses: **LR(0) items // FOLLOW sets**
It is simpler than LALR and CLR.
Steps to Construct SLR Parsing Table
Step 1: Augment the Grammar
Add new start symbol: // S' → S
Step 2: Construct LR(0) Items: Find **closure()** // Find **goto()** // Build **canonical collection of LR(0) items**
Step 3: Construct DFA (State Diagram): Each state = set of LR(0) items // Transitions based on grammar symbols
Step 4: Compute FOLLOW Sets
Used for **reduce actions**
Step 5: Construct Parsing Table
ACTION Table: Shift (S) //Reduce (R) // Accept (acc)
GOTO Table: For non-terminals
Example: Given Grammar:
S → CC // C → cC // C → d
Step 1: Augmented Grammar
S' → S // S → CC // C → cC // C → d
Step 2: LR(0) Items
I_0: // S' → •S // S → •CC // C → •cC // C → •d
I_1 (goto(I_0, S)): S' → S•
I_2 (goto(I_0, C)):// S → C•C // C → •cC // C → •d
I_3 (goto(I_0, c)): C → c•C // C → •cC // C → •d
I_4 (goto(I_0, d)): // C → d•
I_5 (goto(I_2, C))://S → CC• ///**I_6 (goto(I_3, C)): ///** C → cC•

Construct *LALR parsing table*.
LALR (Look-Ahead LR) parser is a type of bottom-up parser obtained by **merging LR(1) states having same LR(0) items**.
It is: // More powerful than **SLR** // More compact than **CLR**
Idea of LALR Parsing
Start with **LR(1) items** // Merge states with **same core (LR(0) items)** // Combine their **lookahead symbols**
Then construct **ACTION and GOTO table**
Steps to Construct LALR Parsing Table
Step 1: Augment the Grammar
Add a new start symbol: // S' → S
Step 2: Construct LR(1) Items : Compute **closure** and **goto** // Build **canonical LR(1) collection**
Step 3: Merge States: Find states with same **LR(0) core** // Merge them // Combine their **lookahead sets**
This reduces number of states
Step 4: Construct Parsing Table
ACTION Table: Shift // Reduce // Accept
GOTO Table: For non-terminals
Example // Given Grammar:
S → L = R // S → R // L → * R // L → id // R → L
Step 1: Augmented Grammar: S' → S
Step 2: LR(1) Items : (Construct full LR(1) states – usually shown in exam briefly) // **Step 3: Merge States**
👉 Example: // If two states have:
L → id • , a // L → id • , b // Merge → /// L → id • , {a, b}

Generate *Three Address Code for control structures*.
Definition of Three Address Code (TAC)
Three Address Code is an **intermediate code representation** in which each instruction contains **at most three addresses**.
General form: // x = y op z
Used in compiler for: Easy optimization // Simple code generation
Control Structures in TAC: If statement // If–Else statement // While loop // Do–While loop
1. TAC for IF Statement: Syntax://if (condition)// S1;
TAC Form: if condition goto L1 // goto L2 // L1: S1 // L2:
Example: if (a < b) // x = y + z;
TAC: if a < b goto L1 // goto L2 // L1: x = y + z // L2:
2. TAC for IF–ELSE Statement
Syntax: if (condition) // S1; // else // S2;
TAC Form: if condition goto L1
goto L2 // L1: S1 // goto L3 // L2: S2 //L3:
Example : if (a > b)// x = a; // else // x = b;
TAC: if a > b goto L1
goto L2 // L1: x = a // goto L3 //L2: x = b // L3:
3. TAC for WHILE Loop: Syntax: while (condition) // S1;
TAC Form: L1: if condition goto L2
goto L3 // L2: S1 // goto L1 // L3:
Example : while (i < 10) // i = i + 1;
TAC: L1: if i < 10 goto L2 // goto L3
L2: i = i + 1 // goto L1 // L3:
4. TAC for DO–WHILE Loop
Syntax: // do {// S1; // } while (condition);
TAC Form: L1: S1 // if condition goto L1
Example: do {// x = x + 1; // } while (x < 5);
TAC: L1: x = x + 1 // if x < 5 goto L1
5. TAC for Nested IF (Important)
Example: if (a > b) // if (c > d) // x = 1;
TAC: if a > b goto L1 // goto L3 // L1: if c > d goto L2 // goto L3 // L2: x = 1 // L3: // **Advantages of TAC:** Simple and easy to understand // Helps in optimization // Machine-independent representation // **Disadvantages:** Large number of instructions Uses many temporary variables

What is the role of parsing? Explain different types of parsing in compiler design.
✅ **Role of Parsing (Syntax Analysis)**
Parsing is the **second phase of a compiler**. It is also called **Syntax Analysis**. 👉 **Definition:** Parsing is the process of analyzing a sequence of tokens to determine whether it follows the **grammar rules (CFG)** of the programming language.
🎯 **Main Roles of Parsing**
1.Checks Syntax Correctness: Ensures program follows language rules.
2.Builds Parse Tree / Syntax Tree: Shows hierarchical structure of program. **3.Detects Syntax Errors:** Missing semicolon, brackets, etc.
4.Provides Structure for Next Phases: Helps semantic analysis and code generation.
Types of Parsing: Parsing is mainly divided into **two types**:
🔷 **1. Top-Down Parsing👉 Definition:** Top-down parsing starts from the **start symbol** and tries to generate the input string.
Direction: Root → Leaves // Types of Top-Down Parsing
(a) Recursive Descent Parsing: Uses **recursive functions**.
>May require **backtracking**. // Simple but less efficient.
(b) Predictive Parsing (LL(1)): >Uses **no backtracking**.
>Uses **FIRST and FOLLOW sets**. >Uses **parsing table**.
Most efficient top-down method.
Advantages: Easy to understand and implement.
Disadvantages: Cannot handle **left recursion**.
Limited grammar support.
🔷 **2. Bottom-Up Parsing👉 Definition:** Bottom-up parsing starts from the **input string** and reduces it to the **start symbol**.
Direction: Leaves → Root // Types of Bottom-Up Parsing
(a) Shift-Reduce Parsing: >Uses **shift and reduce operations**.
Builds parse tree in reverse.
(b) LR Parsing: Most powerful parsing method.
Types: **SLR (Simple) // CLR (Canonical) //LALR (Look Ahead)**
Advantages: Handles **more complex grammars**.
No backtracking required. **Disadvantages:** Difficult to implement.
Parsing tables are large.

Explain *storage allocation strategies* in detail.
Storage allocation strategies refer to the methods used by a compiler to **allocate memory to variables and program data during execution**. It is a part of the **runtime environment**.
Objectives: >Efficient use of memory // >Fast access to variables >Support for recursion and dynamic data
>Proper management of program execution
Types of Storage Allocation Strategies: There are **three main types**: **Static Allocation // Stack Allocation // Heap Allocation**
1. Static Storage Allocation: Memory is allocated **at compile time** and remains fixed throughout program execution.
Features: Fixed memory size // Allocated before execution
Address of variables does not change // **Example:** int x = 10;
Advantages: ✓ Simple and fast // ✓ No runtime overhead
Disadvantages: No support for recursion // No dynamic memory allocation // Wastage of memory
2. Stack Storage Allocation: Memory is allocated and deallocated **at runtime using a stack (LIFO)**. // **Features:** Used for **function calls** // Each function gets an **activation record** // Supports recursion /// **Stack Representation** Top → Function C // Function B // Function A // Bottom // **Example:** void fun() {// int a; // }
Advantages : Efficient memory usage // Supports recursion Automatic allocation and deallocation // **Disadvantages:** Limited memory size // Cannot allocate complex dynamic structures
3. Heap Storage Allocation: Memory is allocated **dynamically at runtime** from the heap. **Features :** Memory allocated as needed // Used for dynamic data structures / **Example :** int *p = (int*)malloc(sizeof(int));
Advantages: Flexible memory usage // Supports dynamic data structures (linked list, tree) // **Disadvantages:** Slower than stack Complex management // Memory leaks possible
Comparison of Storage Allocation Strategies

Feature	Static	Stack	Heap
Allocation Time	Compile time	Runtime	Runtime
Memory Size	Fixed	Dynamic (limited)	Dynamic
Recursion	Not supported	Supported	Supported
Speed	Fast	Fast	Slow
Complexity	Simple	Medium	Complex

Output Tokens:

Lexeme	Token Type
a	Identifier
=	Operator
b	Identifier
+	Operator
10	Number

Role of Symbol Table
Stores information about identifiers:
Name /// Type /// Scope
Lexical Errors
Examples: Invalid symbols (@, # etc.) // Wrong identifiers (1abc)
Handled by: Skipping characters // Error messages
Input Buffering: Used for efficient reading of input // Uses **two-buffer scheme**
LEX Tool: Automatic tool to generate lexical analyzer // Uses regular expressions
Advantages: Simplifies syntax analysis // Improves efficiency // Removes unnecessary data
Disadvantages: Limited error handling // Cannot detect syntax errors

Structure of Activation Record
Return Address → Where control returns after function
Parameters → Input values // **Local Variables** → Variables inside function // **Temporary variables**
Control Link → Points to previous function
Access Link → Used for non-local variables
3. Parameter Passing Methods
Call by Value: Copy of value is passed
✓ Safe // ❌ No modification of original
Call by Reference: Address of variable is passed
✓ Changes reflect // ❌ Risky
4. Access to Variables
Local Variables: Stored in activation record // Accessed directly
Non-local Variables: Accessed using **access link** or **static link**
5. Symbol Table: A data structure used by compiler to store information about variables.
Stores: >Variable name > Type > Scope > Memory location
Data Structures Used: Hash table > Linked list > Tree
6. Heap Management: Handles dynamic memory allocation
Allocates and deallocates memory during runtime
Advantages of Runtime Environment
✓ Efficient memory management // ✓ Supports recursion // ✓ Enables dynamic allocation // ✓ Improves execution control
Disadvantages // Complex implementation
Memory overhead // Risk of memory leaks (heap)

(c) Loop Unrolling: Reduce loop overhead by repeating statements.
6. Copy Propagation: Replace variables with assigned values.
Example: // a = b // c = a → c = b
7. Algebraic Simplification: Simplify expressions using algebra rules. // Example: // x = x + 0 → x = x // x = x * 1 → x = x
8. Peephole Optimization: Looks at small sets of instructions and replaces inefficient ones. Example: // MOV R1, 0 // ADD R1, X → MOV R1, X
2. Machine Dependent Optimization: These depend on hardware architecture. **Techniques**
1. Register Allocation: Efficient use of CPU registers.
2. Instruction Scheduling: Rearranges instructions to improve speed. **3. Use of Special Instructions:** Uses machine-specific instructions for efficiency.
Advantages of Code Optimization : Faster execution
Reduced memory usage // Better CPU utilization
Disadvantages: Increases compilation time
Complex implementation // May increase code size (in some cases)

Step 4: LALR Parsing Table
Final Table Format

State	Id	*	=	$	S	L	R
0	S5		S4		1	2	3
1				acc			

Important Points: LALR merges states → **fewer states than LR(1)** // Same power as LR(1) in most cases
Used in tools like **YACC**
Advantages: Smaller parsing table // Efficient memory usage //Widely used in practice // **Disadvantages:** Complex construction // May introduce conflicts in some cases
Difference: SLR vs LALR vs CLR

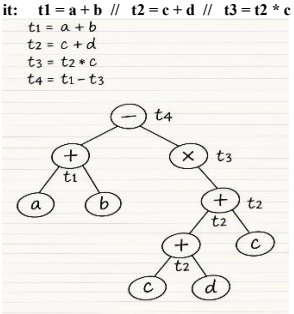
Feature	SLR	LALR	CLR
Power	Low	Medium	High
States	Few	Moderate	Large
Accuracy	Less	More	Highest

Step 3: FOLLOW Sets
FOLLOW(S) = {$} /// FOLLOW(C) = {c, d, $}
Step 4: SLR Parsing Table

State	c	d	$	S	C
0	S3	S4		1	2
1			acc		
2	S3	S4		5	
3	S3	S4		6	
4	R3	R3	R3		
5			R1		
6	R2	R2	R2		

Productions Numbering
1. S → CC // 2. C → cC // 3. C → d
Explanation:>S3, S4 → Shift
R1, R2, R3 → Reduce // > **acc** → Accept
Advantages Simple to construct // Requires less memory
Disadvantages: Less powerful than LR(1) // May produce conflicts

Explanation
Only **one node** is created for (a + a) → reused everywhere
This eliminates **redundant computation**
DAG shares common subexpressions
Optimized Computation: // t1 = a + a // t2 = c * t1
t3 = t2 + t1 // t4 = t1 + t1 /// result = t3 + t4
✅ **Advantages of DAG in this Example**
✓ Avoids recomputing (a + a) multiple times
✓ Reduces number of operations
✓ Improves efficiency
Consider the following basic block and construct the DAG for it: t1 = a + b // t2 = c + d // t3 = t2 * c // t4 = t1 - t3



Explain *syntax-directed translation in detail*.
Syntax-Directed Translation (SDT) is a method in compiler design where **semantic rules (actions)** are attached to the **grammar productions** to perform translation.
It is used to: Generate intermediate code // Build syntax trees // Perform type checking
Basic Idea: >Each grammar rule is associated with **semantic actions** > These actions are executed during parsing > Translation is done **based on syntax structure**
Example: Grammar: // E → E1 + T // E → T
With semantic rule: // E.val = E1.val + T.val
Here, .val is an **attribute**
Components of SDT: 1. Attributes: Attributes store information about grammar symbols. **Types of Attributes: (a) Synthesized Attributes:** Computed from **child nodes** // Flow: **bottom-up**
Example: // E.val = E1.val + T.val
(b) Inherited Attributes: Passed from **parent or sibling nodes**
Flow: **top-down** // Example: // T.type = E.type
2. Syntax-Directed Definition (SDD): An SDD is a **context-free grammar** with **attributes and semantic rules**.
Types of SDD: 1. S-Attributed Definition: Uses only **synthesized attributes** // Evaluated in **bottom-up parsing**
✓ Easy to implement
2. L-Attributed Definition: Uses both **inherited and synthesized attributes** // Evaluated in **top-down parsing** // ✓ More powerful
3. Syntax Tree Construction: SDT helps in building:
>> Parse Tree >>Abstract Syntax Tree (AST)
Example: // E → E + T // Tree: E / \ E T
4. Intermediate Code Generation: SDT is used to generate **Three Address Code (TAC)**. Example: // E → E1 + T
E.addr = newtemp() // E.code = E1.code || T.code ||
E.addr = E1.addr + T.addr
Applications of SDT: Expression evaluation // Type checking Intermediate code generation // Syntax tree construction
Advantages: Systematic translation// Easy to implement with parser// Supports automatic code generation
Disadvantages: Complex for large grammars // Difficult to debug

Q. What is a Compiler? Explain the phases of a compiler with a neat block diagram.
A **compiler** is a system software that translates a program written in a **high-level language (like C, Java)** into **machine-level language (binary code)** so that it can be executed by the computer. It checks for errors and converts the entire program at once.
✅ **Phases of a Compiler:** A compiler works in different steps called **phases**. Each phase performs a specific task in the translation process.
Source Program -->Lexical Analysis
--> Syntax Analysis--> Semantic Analysis--> Intermediate Code Generation --> Code Optimization --> Target Code Generation--> Target Program
Symbol Table (used to store variable info) // **Error Handler** (handles errors in each phase)
1. Lexical Analysis (Scanner): >First phase of compiler.
Converts source code into **tokens**. // Removes spaces, comments.
Example: int a = 10; // → int, a, =, 10
2. Syntax Analysis (Parser)
>Checks whether the program follows **grammar rules**.
>Generates **parse tree**.
Example: // Detects missing semicolon error.
3. Semantic Analysis >Checks **meaning of program**.
>Type checking (int, float, etc.)
Example: // int a; /// a = "hello"; (type error)
4. Intermediate Code Generation
Converts program into **intermediate code (IC)**. // >Machine independent. Example: a = b + c /// → t1 = b + c // → a = t1
5. Code Optimization: Improves code efficiency.
Removes unnecessary instructions. Example: // a = b + 0 → a = b
6. Target Code Generation
Converts optimized code into **machine code**.
Final output is executable program.
Symbol Table: Stores information about variables:
Name // Type // Scope
Error Handling: Detects and reports errors in each phase:
Lexical // Syntax /// Semantic errors

Explain *DAG representation of basic block*.
Definition of Basic Block: A **basic block** is a sequence of consecutive statements in a program with: // **Single entry point** // **Single exit point** // No branching in between
DAG (Directed Acyclic Graph) is a graphical representation used to represent expressions in a **basic block**.
It helps in **code optimization** by identifying:
Common subexpressions // >>Redundant computations
Purpose of DAG: > Eliminate repeated calculations
Optimize expressions // >Improve code efficiency
Structure of DAG: Leaves (Nodes) → Represent variables or constants // > **Interior Nodes** → Represent operators (+, –, *, /) **> Edges** → Show dependency between operands
Example: Given Basic Block: //a = b + c // d = b + c // e = a + d
Step-by-Step DAG Construction: >Create nodes for **b** and **c**
>Create node for **b + c** //>Assign it to **a** and **d** (same node reused)
Create node for **a + d**
DAG Representation (Explanation):>One node represents **b + c**
It is shared by both **a** and **d** // >This avoids recomputation
Final optimized idea: // t1 = b + c // a = t1 // d = t1 // e = t1 + t1
Advantages of DAG: >Eliminates **common subexpressions**
> Reduces redundant calculations >Improves execution speed
Helps in further optimization
Disadvantages: Complex to construct // >Works only within a basic block // > Not suitable for global optimization
Applications of DAG: Code optimization
Compiler design // Expression simplification

Write a lex program to count the number of words, lines, and characters in an input string

```
%{ // #include <stdio.h> /// int words = 0;
int lines = 0; /// int chars = 0; // %} // %%
\n          { lines++; chars++; } // [ \t]+      { chars += yyleng; }
[^ \t\n]+   { words++; chars += yyleng; } // %%
int main() // { //   yylex();
    printf("Number of lines = %d\n", lines);
    printf("Number of words = %d\n", words);
    printf("Number of characters = %d\n", chars);
    return 0; } ///int yywrap() // {  //  return 1; // }
```

Explain top-down and bottom-up parsing techniques in detail. Parsing is the process of analysing a string of tokens to check whether it follows the grammar of a language. It builds a **parse tree** for the given input. There are two main parsing techniques: **1.Top-Down Parsing** // 2.**Bottom-Up Parsing** **1. Top-Down Parsing:** Top-down parsing starts from the **start symbol (root)** and tries to derive the input string step by step. It constructs the parse tree **from root to leaves**. **Working: >**Begins with the **start symbol (S)** >Applies production rules >Expands non-terminals until the input string is obtained **Example:** Grammar: // S → aA // A → b // Input: **ab** Derivation: // S → aA → ab **Types of Top-Down Parsing** **(a) Recursive Descent Parsing:** Uses **recursive procedures** for each non-terminal. > May involve **backtracking**. **Disadvantage:** Time-consuming due to backtracking. **(b) Predictive Parsing (LL(1)):** Uses **no backtracking**. Uses **FIRST and FOLLOW sets**. > Uses **parsing table**. 👉 **Advantage:** Fast and efficient. **Advantages:** Easy to understand and implement Suitable for simple grammars **Disadvantages:** Cannot handle **left recursion** > Limited grammar support // >Backtracking (in some cases)		

		2. Bottom-Up Parsing: Bottom-up parsing starts from the **input string** and reduces it to the **start symbol**. It constructs the parse tree **from leaves to root**. **Working:>**Input string is scanned // > Substrings are reduced to non-terminals //> Continue until only **start symbol** remains **Example //** Grammar: S → aA // A → b Input: **ab //** Steps: // ab → aA → S **Types of Bottom-Up Parsing** 🔹 **(a) Shift-Reduce Parsing:** Uses two operations: **Shift** → Move input to stack **Reduce** → Replace substring using production 🔹 **(b) LR Parsing**:** Most powerful technique. Types: **SLR (Simple LR) // CLR (Canonical LR) //LALR (Look-Ahead LR)** **Advantages:** Handles **complex grammars** No backtracking required // >More powerful than top-down ❌ **Disadvantages:** Difficult to understand Complex implementation // >Large parsing tables 🔁 **Difference Between Top-Down and Bottom-Up Parsing** **Feature** **Top-Down Parsing** **Bottom-Up Parsing** Direction Root → Leaves Leaves → Root Approach Expansion Reduction Start Point Start symbol Input string Backtracking May be required Not required Complexity Simple Complex Example LL(1) LR, SLR